

CAPSTONE PROJECT

YCBS 299 - TEAM 2

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MARCH 2024



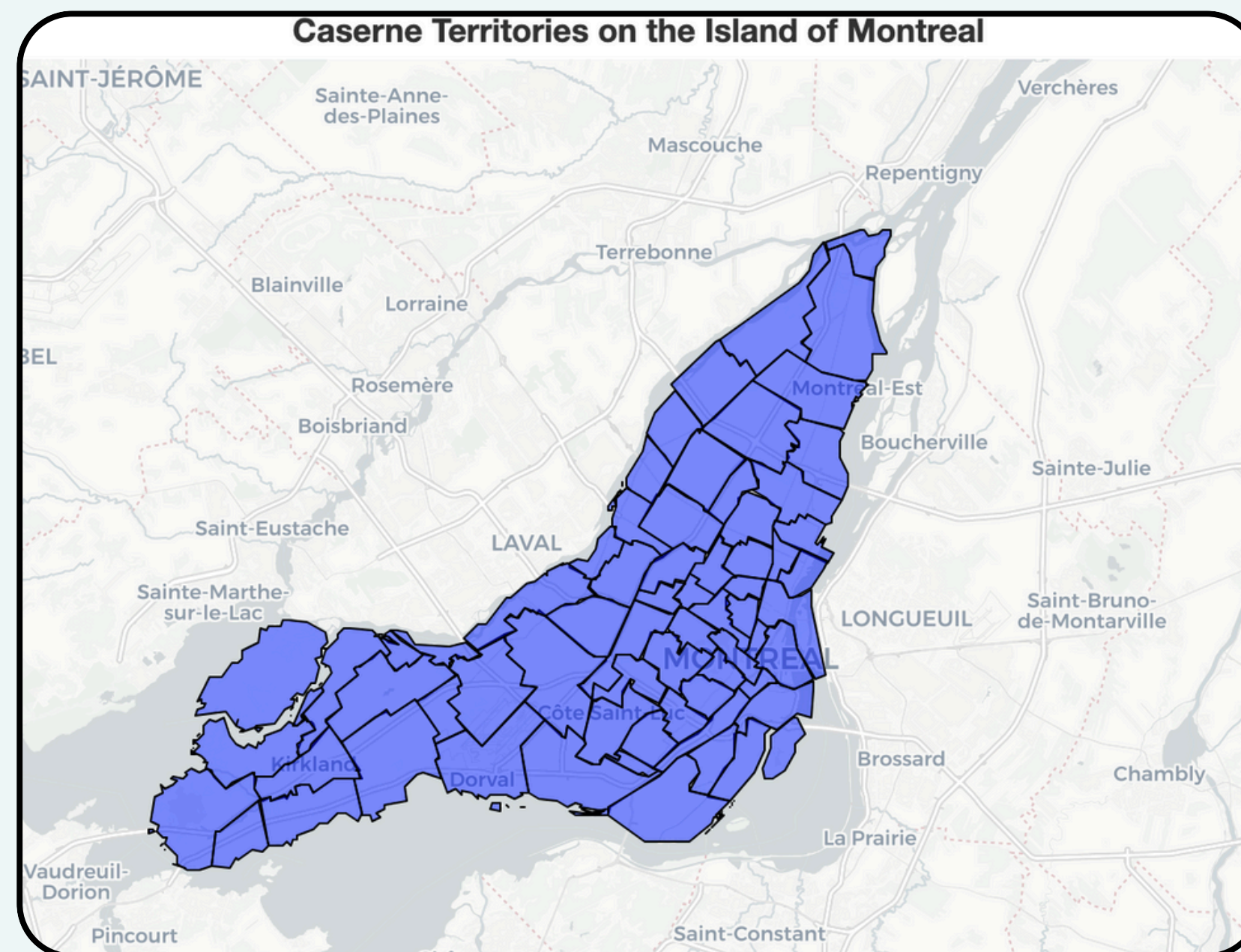
OBJECTIVE

Predict fire risk areas by month in the city of Montreal.



HOW WE DEFINE AREAS ?

Objective : Predict fire risk areas by month in the city of Montreal

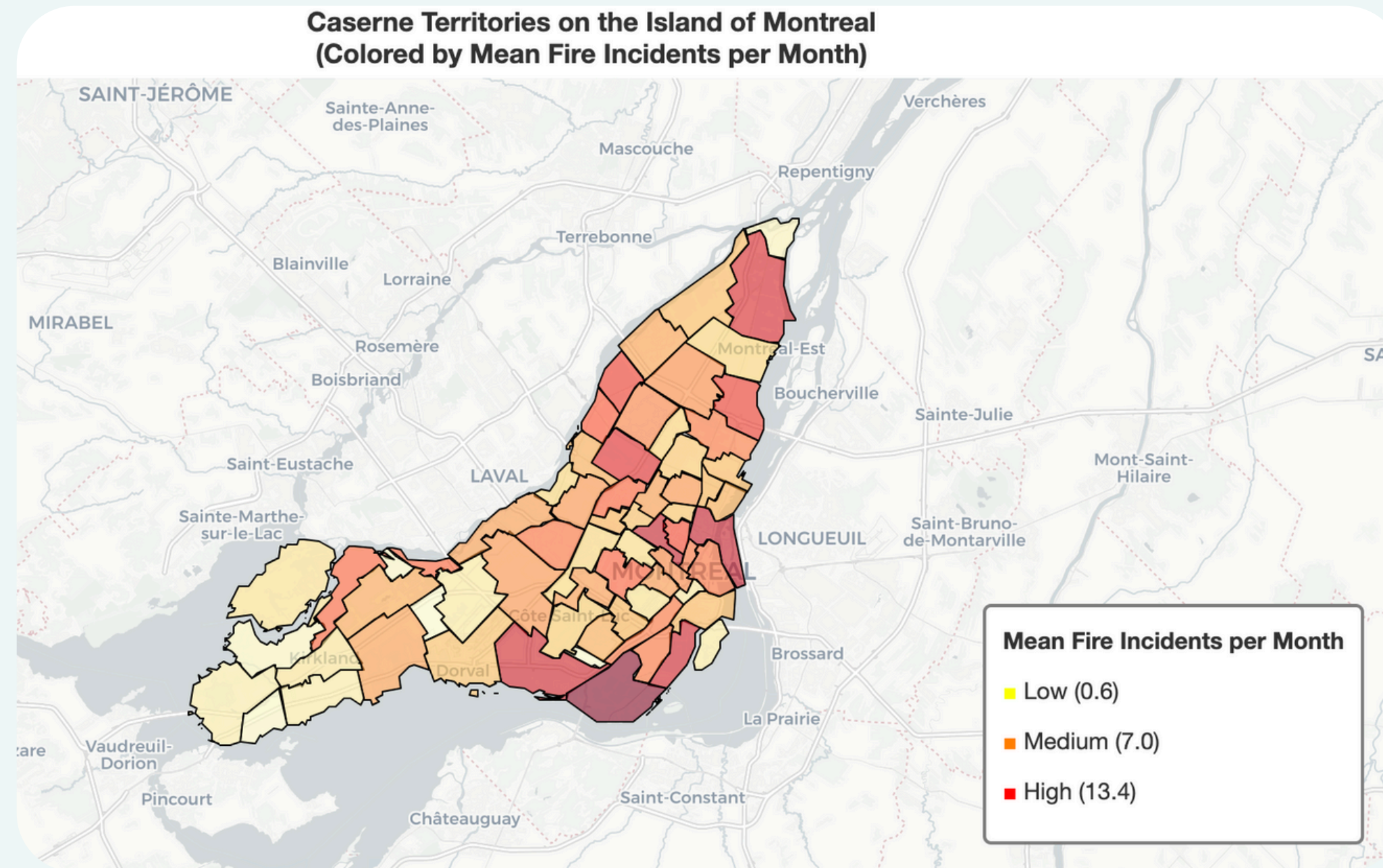


67 administrative caserne areas of the fire stations across the entire Island of Montreal



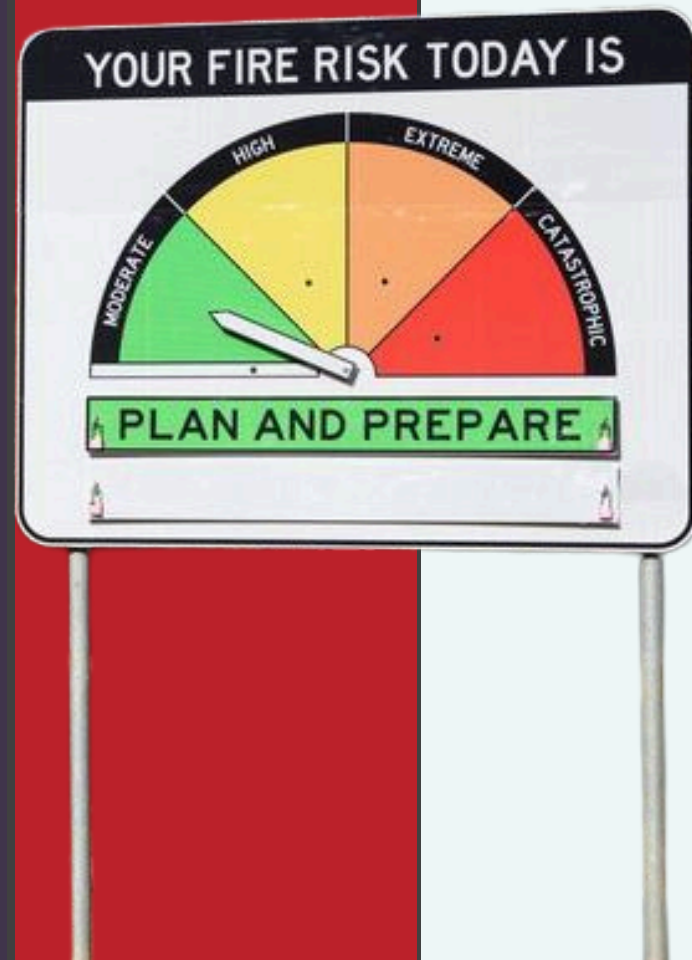
Fire or not in the next 12 months for each caserne territory ? If so, how many ?

We define RISK by the number of fire incidents



WHY ?

Many proactive and actionable insights can be derived from predicting fire risk areas by month in Montreal



BOOST FIRE SAFETY AND PREPAREDNESS THROUGH EDUCATION AND TRAINING

Deploy prevention agents for workshops, fire drills, and smoke detector distribution, while offering more frequent safety training for firefighters on handling building fires.

STRENGTHEN BUILDING CODE COMPLIANCE AND FIRE INSPECTIONS

Conduct compliance audits and regular fire inspections to ensure buildings meet safety codes (e.g., proper exits, fire-resistant materials) and identify potential fire hazards.



ENHANCE FIREFIGHTING AND RESSOURCES SCHEDULING

Better planning for firefighters and specialized equipment during peak fire months to enhance preparedness.

SUPPORT FIREFIGHTER MENTAL HEALTH

Expand mental health programs, including counseling and stress management workshops, for firefighters to address increased stress levels.

Machine learning for fire risk predictions

Fire risk is influenced by numerous factors, including demographic shifts, societal challenges, climate change-driven extreme weather, and urban densification.

ML models can analyze complex patterns across these variables, enabling data-driven decision-making to enhance public safety.

DATA SOURCES

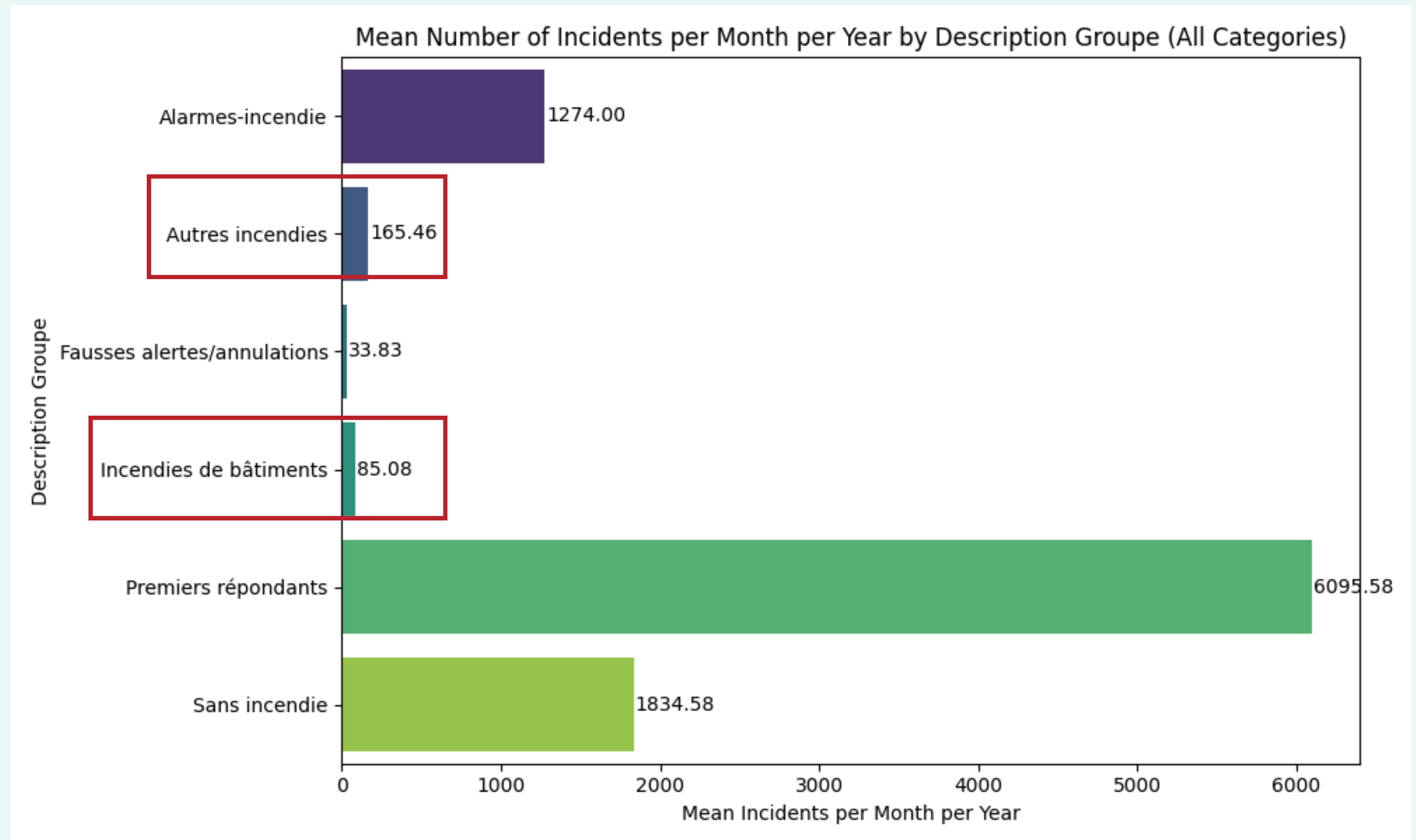
01	Fire Incidents	• Lists of interventions carried out by the Montreal Fire Department (SIM) • https://donnees.montreal.ca/ville-de-montreal/interventions-service-securite-incendie-montreal	Updates : Daily
02	Crimes	• List of criminal offenses recorded by the Montreal Police Service (SPVM). • https://donnees.montreal.ca/ville-de-montreal/interventions-service-securite-incendie-montreal	Updates : Daily
03	Property Assessment	• Montreal property assessment units. • https://donnees.montreal.ca/ville-de-montreal/unites-evaluation-fonciere	Updates : Weekly
04	Weather	• Weather for Montreal • https://open-meteo.com/	Updates : Daily
05	Fire Hydrants	• Fire hydrant locations on the island of montreal • https://donnees.montreal.ca/fr/dataset/geolocalisation-des-bornes-fontaines	Updates : Weekly
06	Territoire admin	• Administrative areas of the fire stations across the entire Island of Montreal • https://donnees.montreal.ca/dataset/territoires-administratifs-des-casernes	Updates : Monthly
07	Population	• Population per borough of the island of Montreal • The gazette, LOIS ET RÈGLEMENTS Partie 2 - Gouvernement du Québec	Updates : Yearly

Total Casernes: 67

- Caserne 79 excluded because it was consolidated in November 2009
- The dataset includes historical data from 2005 to 2024, with consistent features across casernes
- 93 802 actual fire incidents were part of the analysis
- Fire Incidents are aggregated by Caserne, year, month with spatial joins
- Spatial joins were critical for mapping fire incidents, crime data, fire hydrants, and buildings data to the correct caserne territories

FIRE INCIDENTS

☐ Focus of this work is on actual fires



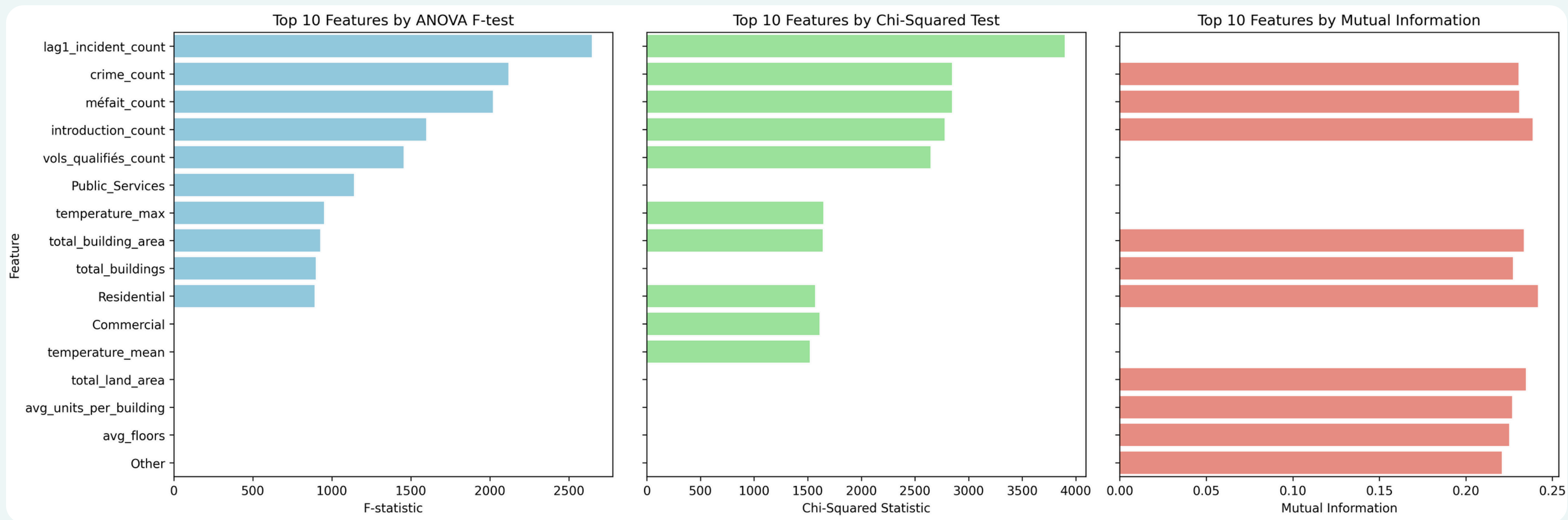
MERGED DATA

caserne	Caserne identifier	avg_units_per_building	Average units per building
year	Year of fire incident	median_construction_year	Median building construction yea
month	Month of fire incident	total_building_area	Total building area (m²)
temperature_max	Max monthly temperature (°C)	total_land_area	Total land area (m²)
temperature_min	Min monthly temperature (°C)	commercial	commercial building count
temperature_mean	Mean monthly temperature (°C)	industrial	industrial building count
precipitation_sum	Total monthly precipitation (mm)	infrastructure	Infrastructure building count
wind_speed_max	Max wind speed (km/h)	other	Other building count
wind_gusts_max	Max wind gusts (km/h)	Public_Services	public service building count
total_buildings	Number of buildings	Recreational	Recreational building count
avg_floors	Average building floors	Residential	Residential building count

MERGED DATA (CONTINUED)

vacant_unused	Vacant/unused building count	Month_Sin	Sine transformation of the month
crime_count	Total crime count	Month_Cos	Cosine transformation of the month
méfait_count	Mischief crime count		
introduction_count	Break-in crime count		
vols_qualifiés_count	Robbery crime count		
hydrant_count	Fire hydrant count		
population	Caserne area population		
area_square_km	Caserne area (km²)		
lag1_incident_count	Previous month's incident		
incident_count	Fire incident count		
risk_level	Risk level (Low/Medium/High)		

FEATURE SELECTION – COMBINED APPROACH



Measures **variance between groups** (Low, Medium, High). Identifies numerical features (e.g., lag1_incident_count) impacting risk levels.

Assesses **independence between categorical features** and target. Highlights features (e.g., crime_count) with strong associations.

Captures **non-linear relationships between features and target**. Ranks features (e.g., total_buildings) by information gain.

FEATURE SELECTION – COMBINED APPROACH

Overlapping Top Features Across Methods				
Feature	temperature_max	1	1	0
	temperature_mean	0	1	0
	total_buildings	1	0	1
	avg_floors	0	0	1
	avg_units_per_building	0	0	1
	total_building_area	1	1	1
	total_land_area	0	0	1
	Commercial	0	1	0
	Other	0	0	1
	Public_Services	1	0	0
	Residential	1	1	1
	crime_count	1	1	1
	méfait_count	1	1	1
	introduction_count	1	1	1
	vols_qualifiés_count	1	1	0
	lag1_incident_count	1	1	0
		ANOVA	Chi-Squared	Mutual Information
Feature Selection Method				

Evaluated features using **ANOVA**, **Chi-Squared**, and **Mutual Information** methods to ensures robust selection by capturing diverse feature relationships. Overlapping top features prioritized for final model.

FINAL FEATURE SELECTION

lag1_incident_count	Selected by all methods; captures historical fire patterns.
crime_count	Selected by all methods; reflects socio-economic risk factors.
total_building_area	Selected by all methods; indicates potential fire impact scale.
total_buildings	High in all methods; reflects structure density.
Residential	Selected by all methods; residential areas have unique risk profiles
temperature_max	High in ANOVA and Chi-Squared; accounts for seasonal risk variations.
Public_Services	Selected by ANOVA; public buildings (e.g., hospitals) influence risk.
avg_floors	High in Mutual Information; taller buildings pose greater fire challenges.

Notes

- Excluded redundant features (e.g., crime subcategories) due to collinearity.
- Features like méfait_count, introduction_count overlapped but were dropped due to collinearity with crime_count.

METHODS EVALUATED

- **Random Forest:** Robust to imbalanced data, reduces overfitting, and highlights key features.
- **XGBoost:** Boosting technique improves accuracy while built-in regularization prevents overfitting.
- **RNN:** a neural network for modeling time series data, using internal memory to retain past information for current processing.

Methods studied and used in other similar projects such as the Firebird: Predicting Fire Risk and Prioritizing Fire Inspections in Atlanta

Classification

- High Risk (Top 20%)
- Medium Risk
- Low Risk (Bottom 50%)

Random Forest

XGBoost

RNN

Regression

Number of Fire
Incidents

Random Forest

XGBoost

RNN

Classification

XGBoost

Accuracy: 0.68



Random Forest

Accuracy: 0.66

RNN

Accuracy: 0.654

Regression

XGBoost

MSE: 9.777

Random Forest

MSE: 10.36

RNN

MSE: 9.408



FINAL MODEL SELECTION

Classification

By defining clear risk categories, it provides an intuitive framework for fire departments to prioritize inspections, allocate resources, and implement targeted fire prevention strategies.



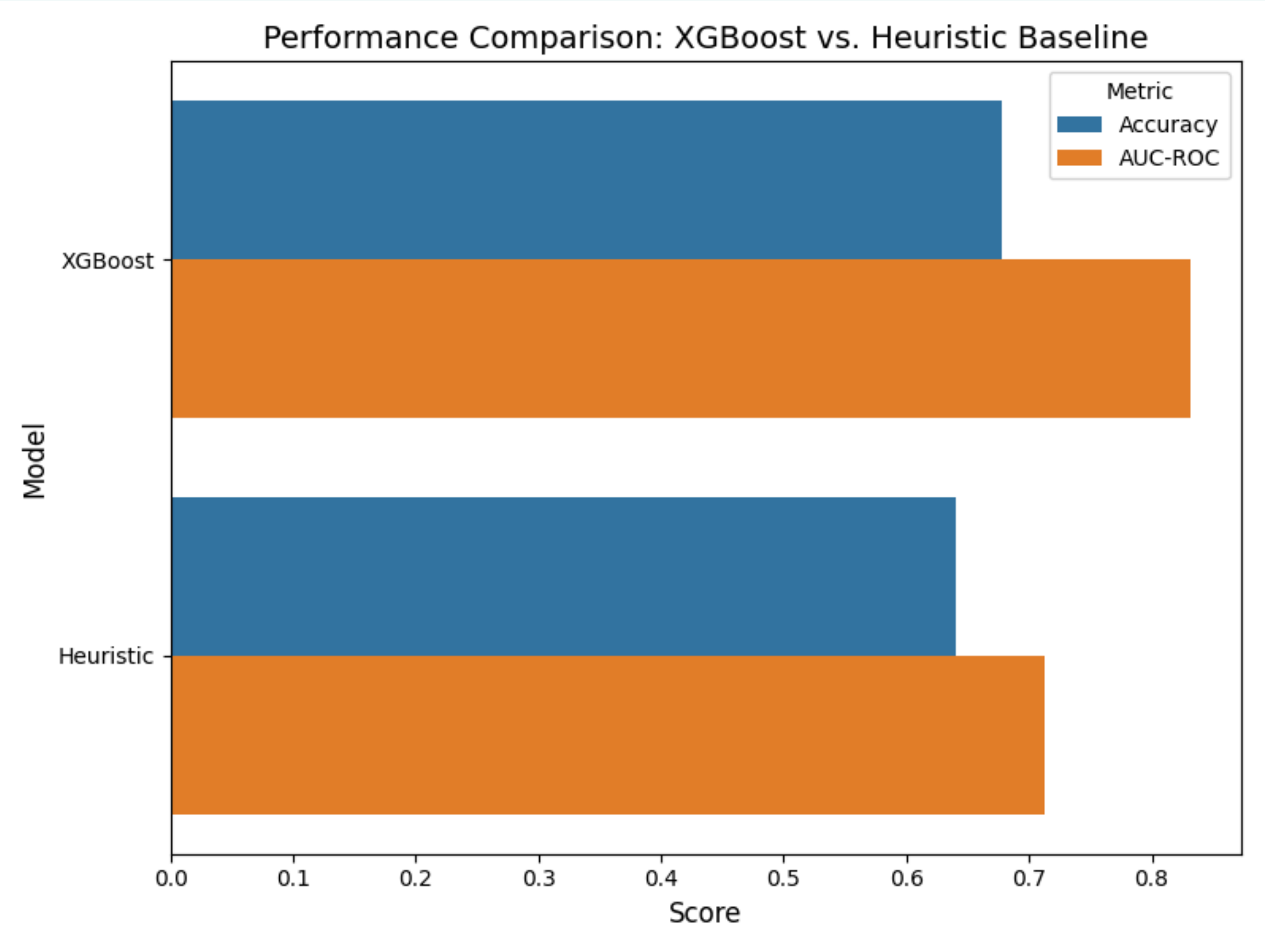
XGBoost

Balances predictive performance with interpretability, making it ideal for planning and actionable decision-making.



Very fast to run (less than a minute)

IS MACHINE LEARNING NECESSARY ?

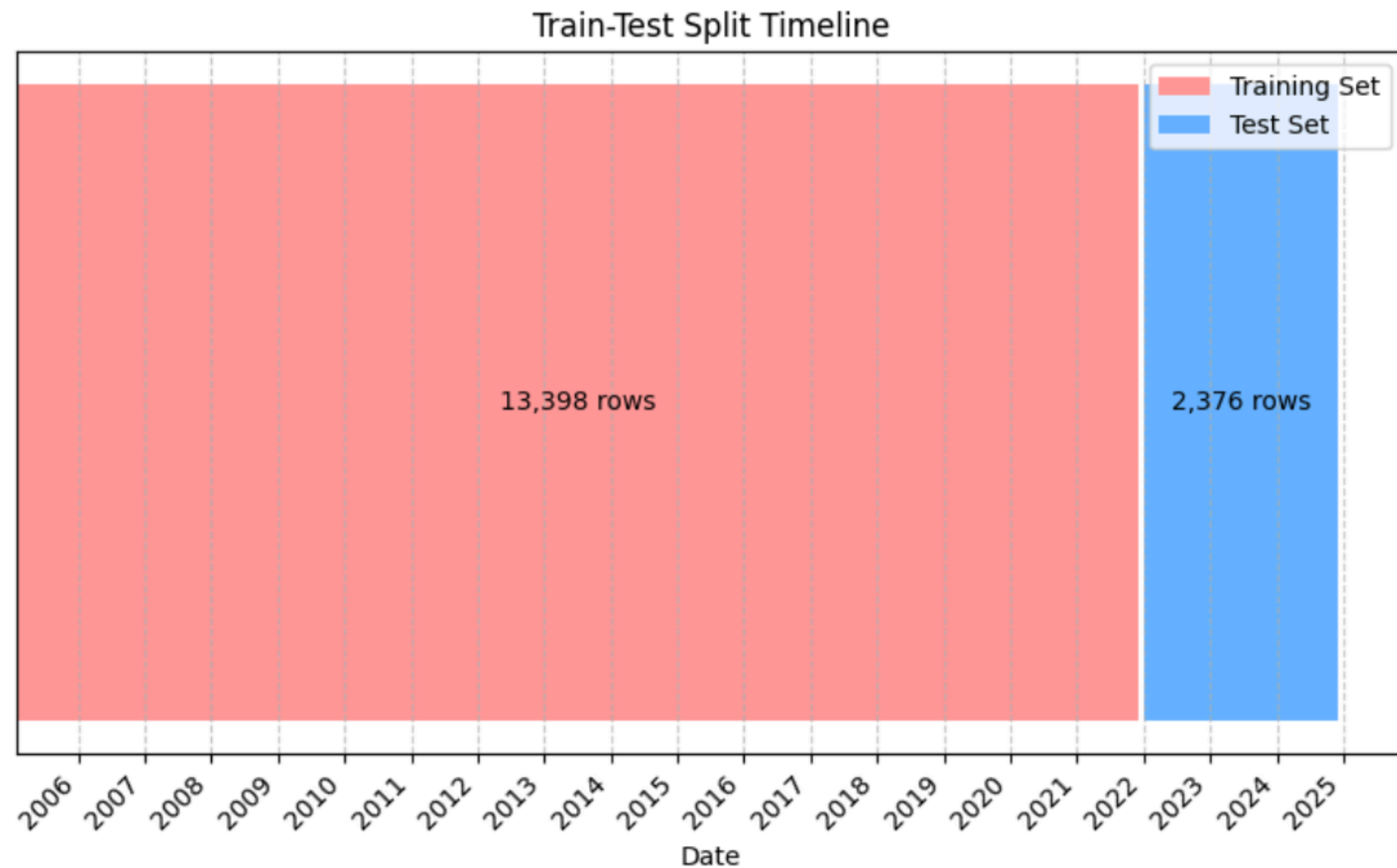


Machine learning outperforms heuristics by capturing complex risk factors, leading to higher accuracy (**68% vs. 64%**) and better AUC-ROC (**0.831 vs. 0.713**).

Even a **4%** accuracy gain means **hundreds of better predictions**, improving resource allocation and reducing missed high-risk areas. Unlike static heuristics, **ML adapts** and **improves** over time, making fire risk prediction more precise, scalable, and **life-saving**.

*Heuristic computed the historical mean of fire incidents for each caserne per month, assigned risk levels, and used these to predict risks for the test set, which was then evaluated and compared against XGBoost

CLASSIFICATION

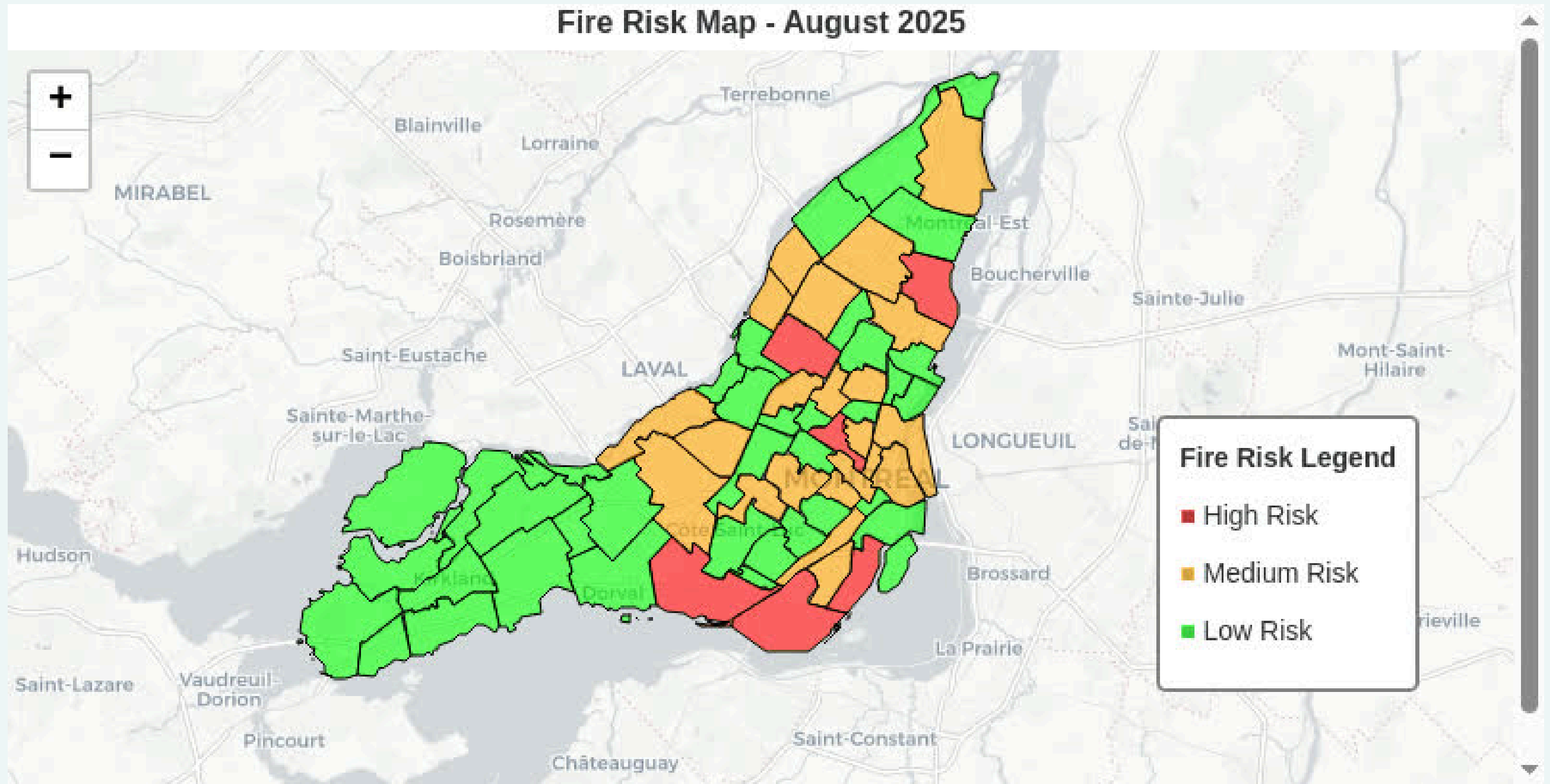


Best Parameters:
Learning Rate: 0.1
Max Depth: 6
N Estimators: 100

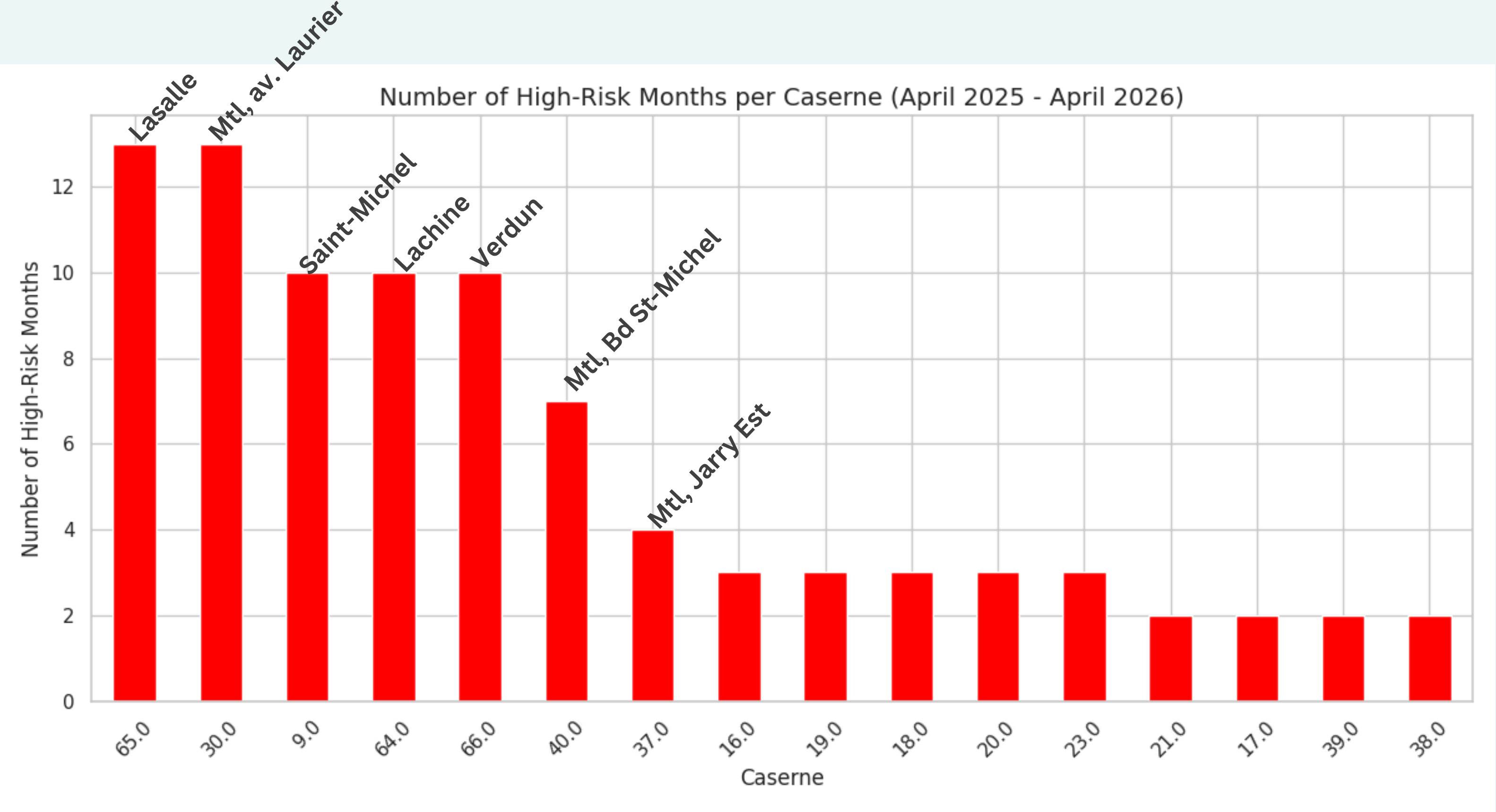
Best Cross-Validated Accuracy: 0.680

Predictions generated as of March 19th, forecasting fire risk levels for the next 12 months.

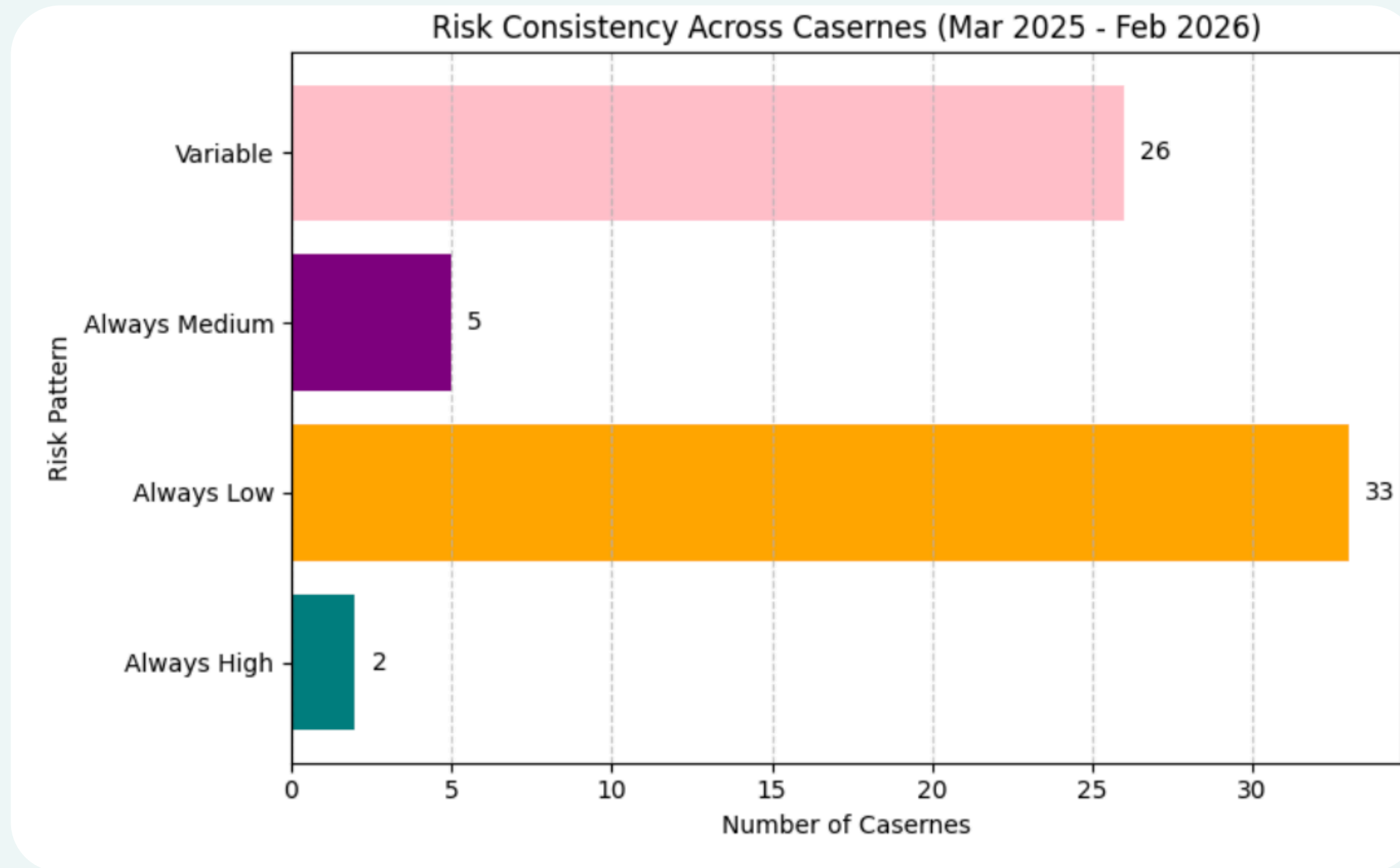
Animation for predictions of April 2025 to April 2026



CASERNE RISK



FIRE RISK PREDICTIONS



Thank You ! Questions?

